University Library Management System

Assignment 1

Project planning and GitHub

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Version 1.0 | Prepared 12 November 2026 | Asia/Karachi

Proposed planning window: 13 November–14 December 2026. Official submission deadlines and university/department details have not been supplied. All later phases are planned only.

112 base person-hours · 16 work packages · 12 risks
One named student; three functional roles

Repository: https://github.com/talalahmad-ai/university-library-management-system.

The repository has a literal final period, unlike the exact name in the PDF. Owner/instructor confirmation is required before submission. Prepared with Codex assistance; no student participation, interviews or testing results are fabricated.

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The Markdown documents and planning-data.json are the editable sources. XLSX formulas include cached results. Standalone vector Gantt/WBS PDFs and PNGs are supplied for readable viewing at full size. The evidence index and compliance matrix record final remote verification separately.

1. Project definition and scope

The case scenario describes a university library using manual or outdated records for borrowing, returns, membership and overdue books. The academic project analyzes and designs an LMS; complete implementation is not required. Phase 1 produces planning and GitHub evidence.

Capability	Planned analysis/design boundary
Catalog and copies	Searchable books and copy availability; no live inventory import.
Membership	Student/faculty membership and librarian/admin roles; policies unconfirmed.
Borrowing and returns	Member validation, available-copy loans, due dates, return and overdue handling.
Reservations	Search/detail/reserve interaction; queue precedence and expiry need confirmation.
Overdue notifications	Notification information and triggers as design concepts; delivery rules unconfirmed.

Excluded: production backend/frontend, live messaging, payment processing, SSO, hosting, real record migration and fabricated human evidence. Success is a coherent, traceable and reviewable coursework package. See reports/project-charter.md for full assumptions, governance and acceptance measures. Confirmed student: Talal Ahmad. All planned packages are assigned to him. Hamza-Khaalid is the authenticated helper; preparation commits are attributed to Codex. University/department and official deadlines remain unknown. Capacity of six hours per calendar day is an estimate requiring student confirmation.

Gate	Planned measurable outcome
A1	Planning artifacts, README/folders, assigned task Issues, at least three meaningful new remote commits and genuine screenshots.
A2 — future	Stakeholder analysis, required FR/NFR/stories/criteria, AI-review decisions and structured SRS/traceability.
A3 — future	Four editable UML models with exports/explanations consistent with SRS.
A4 — future	Five connected screens, two workflows, real feedback from two classmates and two evidenced improvements.

2. Work Breakdown Structure

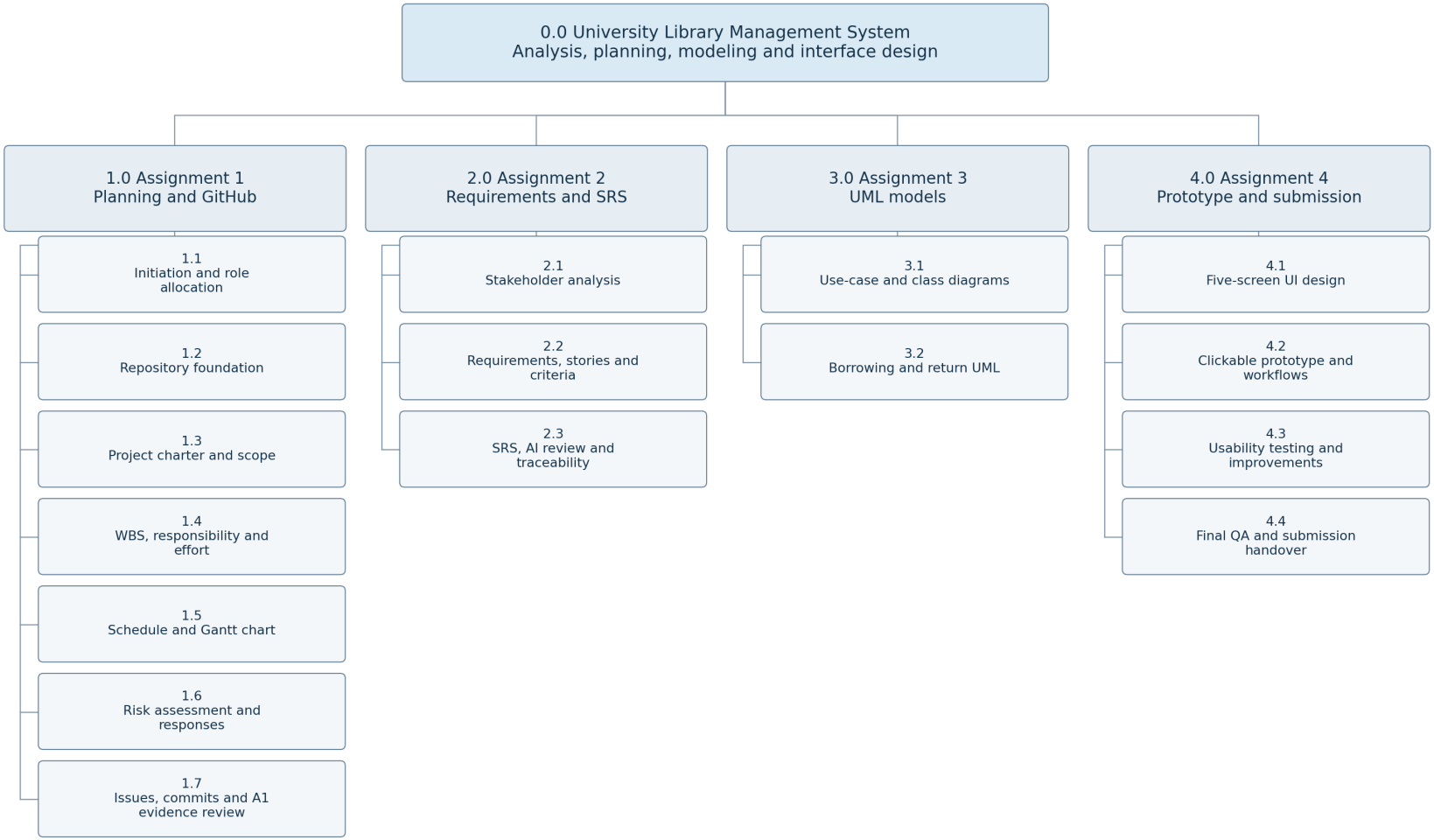
Root 0.0 contains four assignment branches. The following 16 leaf packages are effort-bearing; summary branches and milestone gates add no duplicate effort. Editable diagram: reports/wbs.drawio.

ID	Package	Deliverable
1.1	Initiation and role allocation	Source review, named student and role allocation
1.2	Repository foundation	README, four tracked folders and preserved remote history
1.3	Project charter and scope	Charter with objectives, boundaries, assumptions and success criteria
1.4	WBS, responsibility and effort	WBS source/export, named RACI and task/person effort totals
1.5	Schedule and Gantt chart	Editable schedule and dated Gantt with milestones
1.6	Risk assessment and responses	Scored risk register with triggers, mitigations and contingencies
1.7	Issues, commits and A1 evidence review	Assigned Issues, genuine remote commits, screenshots and A1 audit
2.1	Stakeholder analysis	Stakeholder-needs table and policy questions
2.2	Requirements, stories and criteria	Requirements list, user stories and measurable acceptance criteria
2.3	SRS, AI review and traceability	Structured SRS, AI-review decisions and traceability matrix
3.1	Use-case and class diagrams	Editable use-case/class diagrams, exports and explanations
3.2	Borrowing and return UML	Borrowing sequence and return/overdue activity diagrams
4.1	Five-screen UI design	Login, student dashboard, search, details and librarian screens
4.2	Clickable prototype and workflows	Figma prototype with reservation and issue/return workflows
4.3	Usability testing and improvements	Two real classmates' feedback and two evidenced improvements
4.4	Final QA and submission handover	Cross-artifact QA, evidence index and final submission package

3. WBS visual

University LMS | Work Breakdown Structure

16 effort-bearing packages · Talal Ahmad · Phase 1 execution; Assignments 2–4 planned only



Editable sources: reports/wbs.drawio · reports/wbs.csv · reports/planning-data.json. Summary branches and milestones carry no extra effort.

4. Schedule and dependencies

Proposed calendar-day plan, including weekends. Every dependency is finish-to-start: a successor begins after its predecessor ends. Dates are targets, not official deadlines or actual work records. All tasks are assigned to Talal.

ID	Activity	Start → End (2026)	Days	Depends on	Hours
1.1	Initiation and role allocation	13 Nov → 13 Nov	1	None	4
1.2	Repository foundation	14 Nov → 14 Nov	1	1.1	3
1.3	Project charter and scope	14 Nov → 15 Nov	2	1.1	6
1.4	WBS, responsibility and effort	16 Nov → 16 Nov	1	1.3	6
1.5	Schedule and Gantt chart	17 Nov → 17 Nov	1	1.4	5
1.6	Risk assessment and responses	18 Nov → 18 Nov	1	1.5	4
1.7	Issues, commits and A1 evidence review	19 Nov → 20 Nov	2	1.2, 1.5, 1.6	6
2.1	Stakeholder analysis	21 Nov → 23 Nov	3	1.7	7
2.2	Requirements, stories and criteria	24 Nov → 26 Nov	3	2.1	12
2.3	SRS, AI review and traceability	27 Nov → 29 Nov	3	2.2	12
3.1	Use-case and class diagrams	30 Nov → 02 Dec	3	2.3	10
3.2	Borrowing and return UML	03 Dec → 05 Dec	3	3.1	8
4.1	Five-screen UI design	30 Nov → 03 Dec	4	2.3	8
4.2	Clickable prototype and workflows	06 Dec → 08 Dec	3	3.2, 4.1	10
4.3	Usability testing and improvements	09 Dec → 11 Dec	3	4.2	6
4.4	Final QA and submission handover	12 Dec → 14 Dec	3	2.3, 3.2, 4.3	5

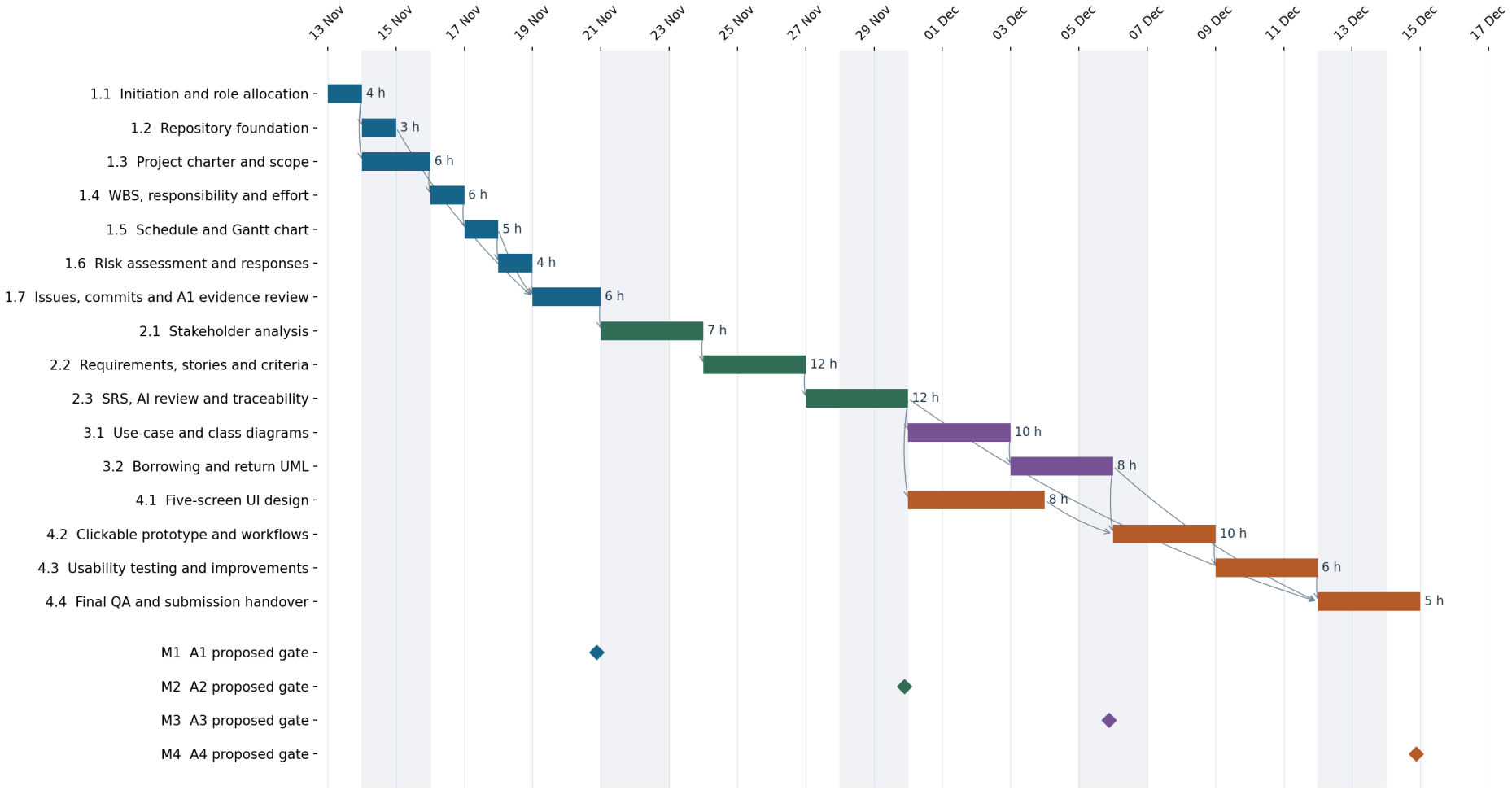
Gate	Proposed target	Official deadline
M1 / A1	2026-11-20	Not supplied
M2 / A2	2026-11-29	Not supplied
M3 / A3	2026-12-05	Not supplied
M4 / A4	2026-12-14	Not supplied

Critical dependency chain: 1.1 → 1.3 → 1.4 → 1.5 → 1.6 → 1.7 → 2.1 → 2.2 → 2.3 → 3.1 → 3.2 → 4.2 → 4.3 → 4.4, 32 calendar days. Task 1.2 has four days of float; task 4.1 has two. Rebaseline when real deadlines or availability are known.

5. Proposed Gantt chart

University LMS | Proposed project schedule

13 November–14 December 2026 • One student: Talal Ahmad • 112 base person-hours



PROPOSED DATES — official deadlines not supplied. Calendar days include weekends. Arrows: finish-to-start; diamonds: end-of-day gates.
A1 blue · A2 green · A3 purple · A4 orange. Later phases are planned only. The schedule is not a record of actual work dates.

6. Responsibility and effort

Talal Ahmad (@talalahmad-ai) is A/R on every row: Accountable for acceptance and Responsible for execution. No additional student is named for Consulted or Informed duties. Functional role allocations describe one person’s types of work, not three people. Estimates are not actual hours or evidence of participation.

ID	Functional lead	Talal RACI	Lead h	Analyst h	Designer h	Talal total h
1.1	Lead	A/R	2	1	1	4
1.2	Lead	A/R	2	0.5	0.5	3.0
1.3	Analyst	A/R	1	4	1	6
1.4	Lead	A/R	4	1	1	6
1.5	Lead	A/R	3	1	1	5
1.6	Designer	A/R	1	1	2	4
1.7	Lead	A/R	4	1	1	6
2.1	Analyst	A/R	1	5	1	7
2.2	Analyst	A/R	2	8	2	12
2.3	Analyst	A/R	2	8	2	12
3.1	Designer	A/R	1	2	7	10
3.2	Designer	A/R	1	1	6	8
4.1	Designer	A/R	1	1	6	8
4.2	Designer	A/R	1	1	8	10
4.3	Designer	A/R	1	1	4	6
4.4	Lead	A/R	3	1	1	5
TOTAL	One student		30	37.5	44.5	112

Phase totals: A1 34 h; A2 31 h; A3 18 h; A4 29 h. Base = 112 h. Unallocated 15% estimate reserve = 16.8 h; envelope = 128.8 h. Reserve is not scheduled task effort or guaranteed calendar float. Combined daily load uses every concurrent task across all roles and does not exceed the assumed six-hour capacity; the overlapping UML/UI tasks peak below that cap.

Review actuals at each gate, reassess any >20% variance and revise dates when capacity is lower or a trigger occurs. sources: effort-estimates.csv, responsibility-matrix.csv and workbook Effort/Capacity sheets.

7. Risks, mitigation and contingency

Likelihood: 1 rare, 2 unlikely, 3 possible, 4 likely, 5 almost certain. Impact: 1 negligible, 2 minor rework, 3 material task delay, 4 phase-gate threat, 5 mandatory submission at risk. Score = L × I. Low 1–5; Medium 6–14; High 15–25. These are qualitative judgments. Talal owns every risk. Review high risks each work session and all risks at every gate.

Highest priorities: R03 effective write access failure (25, currently triggered by HTTP 403), R01 ambiguous policies and R05 unknown official deadlines (16 each). The initial URL issue was resolved; publishing and Issue creation remain blocked. The repository-name exception remains a disclosed decision.

R01 — Ambiguous lending and reservation policies

Likelihood 4 · Impact 4 · Score 16 · Monitoring role Analyst · Owner Talal Ahmad

Trigger: Conflicting or unanswered policy questions at SRS review

Mitigation: Maintain a policy-question log; prioritize librarian confirmation before baselining.

Contingency: Record explicitly proposed rules, isolate affected requirements and re-review after clarification.

R02 — Effort estimates exceed one student's availability

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Lead · Owner Talal Ahmad

Trigger: Actual hours exceed estimate by 20% or a planned day exceeds six hours

Mitigation: Allocate all role hours to Talal; check aggregate daily demand; review estimates at each gate.

Contingency: Use the 15% effort reserve, re-sequence discretionary polish and propose revised dates to the instructor.

R03 — Repository URL or effective write access fails

Likelihood 5 · Impact 5 · Score 25 · Monitoring role Lead · Owner Talal Ahmad

Trigger: Push, Git data write or Issue creation returns an access error; currently triggered by HTTP 403

Mitigation: Use the verified trailing-dot URL; verify effective Contents and Issues write access rather than repository role alone; retain the local Git bundle.

Contingency: Owner restores app access or publishes from an authorized CLI account; retain local commits and package without creating a replacement remote.

R04 — Student unavailable during a critical work package

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Lead · Owner Talal Ahmad

Trigger: Two planned work sessions missed or a milestone is forecast late

Mitigation: Keep tasks small and editable; confirm calendar availability; preserve handover notes.

Contingency: Replan around remaining capacity; request a deadline adjustment if needed; do not invent another contributor.

R05 — Official deadlines differ from proposed dates

Likelihood 4 · Impact 4 · Score 16 · Monitoring role Lead · Owner Talal Ahmad

Trigger: Instructor supplies dates earlier than current proposed gates

Mitigation: Keep every proposed date labeled; request actual submission dates before baselining.

Contingency: Recalculate the dependency network and effort/capacity; obtain revised scope or timing agreement.

R06 — Stakeholders or two classmates unavailable

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Analyst · Owner Talal Ahmad
Trigger: No consultation or testing slot confirmed before the relevant phase
Mitigation: Request availability early and prepare focused questions and a short test script in the appropriate phase.
Contingency: Reschedule with actual eligible participants; disclose missing evidence and seek instructor guidance.

R07 — Git conflicts or accidental loss of existing work

Likelihood 2 · Impact 4 · Score 8 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Remote head changes, conflict appears, or an export is overwritten
Mitigation: Inspect before editing; fetch before pushing; keep editable sources; use normal pushes and reviews.
Contingency: Resolve on a branch, recover from Git and verify against the last reviewed baseline; never force-push.

R08 — Figma access or prototype interactions delayed

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Designer · Owner Talal Ahmad
Trigger: Figma creation, sharing or interaction checks fail at UI kickoff
Mitigation: Confirm account/tool access before design; retain source assets and a workflow checklist.
Contingency: Prepare a local aid and precise Figma completion steps; mark the Figma requirement incomplete until the actual file works.

R09 — SRS, UML and UI use inconsistent rules or terms

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Analyst · Owner Talal Ahmad
Trigger: Review finds a workflow or term that cannot be traced to the baseline
Mitigation: Use a glossary, requirement IDs and cross-artifact review at M2–M4.
Contingency: Pause dependent approvals, fix the source requirement and update all impacted diagrams/screens.

R10 — Testing or screenshots are missing or unreadable

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Designer · Owner Talal Ahmad
Trigger: Evidence index contains a missing file, illegible image or unsupported claim
Mitigation: Capture real pages/artifacts, record URLs/times, open exports and audit every evidence link.
Contingency: Recapture actual state; mark unavailable evidence Pending or Blocked rather than recreating it.

R11 — Scope expands into production implementation

Likelihood 3 · Impact 3 · Score 9 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Tasks add backend, deployment, real data migration or live notifications
Mitigation: Keep the PDF's analysis/design boundary in the charter and review every scope change.
Contingency: Defer implementation requests and restore effort to mandatory assessment artifacts.

R12 — Repository trailing period conflicts with required literal name

Likelihood 3 · Impact 3 · Score 9 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Instructor requires the exact repository spelling from the PDF
Mitigation: Disclose the name difference and ask the owner/instructor to confirm acceptability.
Contingency: Owner renames the existing repository; update origin, document links, Issue map and screenshots; preserve history.

8. GitHub evidence and final review

The exact trailing-dot repository was cloned and inspected before integration. Initial commit d6eb076 and its README are preserved in local history. Initial metadata reported Write permission, an unprotected main branch, no rulesets and zero Issues. talalahmad-ai is an eligible assignee. Effective Git push and the connector Git-data write API both returned HTTP 403. The remote is unchanged: one initial commit, one README and zero Issues. No replacement repository was created.
Genuine incremental preparation commits use the Codex identity and current timestamps. Issue definitions name Talal as the planned assignee. Initial live Issue creation returned HTTP 403, Resource not accessible by integration; do not treat drafts as real Issues. Consult the final evidence index for any later permission resolution and actual counts. Later-phase Issues, if created, do not mean those assignments were executed. Final live counts, commit SHAs, URLs and capture times are recorded in reports/evidence/remote-verification.json and capture-manifest.json after publication.

Submission item	Where to inspect
Repository URL / README	README.md; actual GitHub root screenshot
Remote commit history	reports/evidence/remote-verification.json and GitHub commits screenshot
Assigned task Issues	reports/issue-map.md and genuine Issues/individual-Issue screenshots
Gantt / schedule	project-schedule.csv, assignment-1-planning.xlsx, gantt-chart.pdf / PNG
WBS / responsibilities	wbs.drawio / CSV / PDF; responsibility-matrix.csv; effort-estimates.csv
Risk assessment	risk-register.md / CSV and workbook Risks sheet
Final audit	compliance-matrix.md; quality-review.md; evidence/README.md

Before submission: obtain official deadlines, confirm Talal’s availability and resolve whether the repository’s final period is permitted. The owner already controls the existing repository; no ownership transfer is needed. Do not begin Assignment 2 without the user’s instruction.
Revision 1.0 — 12 November 2026. Sources: instructor PDF (Assignment 1, page 1); uploaded README, starter ZIP and pasted text (supporting context); user-confirmed student and corrected URL. The active request limits execution to Phase 1.